Sales Order Import & Retraining

Full Code Architecture & Implementation Reference Manual

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1. Architecture Overview

The Sales Order Importation and Machine Learning Retraining feature is implemented as a synchronized pipeline spanning the Angular frontend dashboard, the FastAPI backend server, and the background machine learning trainers. The user workflow progresses through validation, confirmation, execution, and dynamic retraining without introducing any application downtime or event-loop blockage.

> ***Zero-Downtime Guarantee:*** *During the 15-second model training period, the backend remains responsive to incoming requests. Stale weights are retained in memory and hot-swapped atomically once training succeeds.*

Pipeline Workflow Stages:

1. **Selection & Column Validation:** User uploads a file (.csv or .xlsx). The frontend posts the file headers to the backend to verify that all required features exist.

2. **Verification Modal:** If valid, a dialog shows mapped columns and the total number of orders. Clicking 'Import' triggers execution.

3. **Background Ingestion:** The backend ingests the sales orders, deduplicates columns to prevent pandas Series mapping errors, upserts records to MongoDB, and registers a FastAPI BackgroundTasks trigger.

4. **Asynchronous Training Loop:** The backend yields the event loop using asyncio.create_subprocess_exec to run python scripts for LightGBM, KNN, KMeans, and ARIMA in a non-blocking subprocess.

5. **Hot-Reloading:** New weights are saved atomically to temporary pickles, then swapped. Active inference routes reload the updated models dynamically.

2. Frontend UI Implementation

The frontend is written in Angular. We replaced native browser prompts with custom, unified HTML modal overlays. Below is the TypeScript controller implementation that validates files, executes imports, and handles real-time retraining progress polling.

2.1 File Upload & Execution Logic

Located in sales-order.ts, this code handles column validation requests and posts imports:

```
validateColumns(): void {
  if (!this.selectedFileForImport) return;
  const token = this.auth.getToken();
  if (!token) return;

  const formData = new FormData();
  formData.append('file', this.selectedFileForImport);

  const headers = new HttpHeaders({
    'Authorization': `Bearer ${token}`
  });

  this.http.post<any>('http://127.0.0.1:8000/api/orders/validate', formData, { headers }).subscribe({
    next: (res) => {
      if (res.status === 'success') {
        this.mappedColumns = res.columns;
        this.importOrderCount = res.order_count;
        this.isImportModalOpen = false;
```

```
        this.isConfirmModalOpen = true;
      } else {
        this.isImportModalOpen = false;
        this.validationErrorMessage = res.message;
        this.isValidationErrorModalOpen = true;
      }
      this.cdr.detectChanges();
    },
    error: (err) => {
      this.isImportModalOpen = false;
      this.validationErrorMessage = "Server error occurred during validation.";
      this.isValidationErrorModalOpen = true;
      this.cdr.detectChanges();
    }
  });
}
```

2.2 Modal Dismissibility during Active Retraining

To let the user navigate the app while training finishes, the close buttons in the progress modal are fully enabled and do not block interface interactions:

```
closeRetrainStatusModal(): void {
  this.isRetrainStatusModalOpen = false;
  this.cdr.detectChanges();
}
```

3. Backend Ingestion & Deduplication

The backend routes are implemented in FastAPI. When a spreadsheet is imported, columns are mapped. However, raw datasets often contain duplicate mappings (e.g. Customer Id and Order Customer Id both mapped to customer_id). We resolve this with pandas deduplication to prevent Series casting crashes.

3.1 API Route Ingestion Code

Located in BackEnd/app/routers/orders.py, this endpoint deduplicates columns and processes rows:

```
@router.post("/import", status_code=status.HTTP_202_ACCEPTED)
async def import_orders_endpoint(
    background_tasks: BackgroundTasks,
    file: UploadFile = File(...),
    db = Depends(get_db),
    current_admin: dict = Depends(get_current_admin)
):
    # ... read file and define col_mapping ...
    df = df.rename(columns=rename_dict)

    # CRITICAL: Drop duplicate renamed columns
    # This prevents group extraction from evaluating as a Series
    df = df.loc[:, ~df.columns.duplicated()]

    # ... validate and group by order_id ...
    for order_id, group in grouped:
        first_row = group.iloc[0]
        cust_id = str(int(first_row["customer_id"]))
        # ... build document and upsert to MongoDB ...
        await db["sales_orders"].update_one(
            {"id": int(order_id)},
            {"$set": order_doc},
            upsert=True
        )

    background_tasks.add_task(retrain_all_models_task, db)
    return {"message": "Success"}
```

4. Asynchronous Retraining & ARIMA Forecasting

Once orders are ingested, models are retrained. To keep the event loop responsive, we run scripts in async subprocesses. We use ARIMA to forecast demand, adding reproducible seeded noise to reflect natural daily swings instead of outputting a flat expectation line.

4.1 Asynchronous Task Execution

In ml_update.py, we execute scripts asynchronously:

```
async def run_subprocess_async(args: list) -> bool:
    try:
        process = await asyncio.create_subprocess_exec(
            *args,
            stdout=asyncio.subprocess.PIPE,
            stderr=asyncio.subprocess.PIPE
        )
        stdout, stderr = await process.communicate()
        return process.returncode == 0
    except Exception:
        return False
```

4.2 ARIMA Forecasting Service

In forecast_service.py, we fit the ARIMA model and generate forecasts with seeded noise:

```
model = ARIMA(df_ts["y"], order=(1, 1, 1), seasonal_order=seasonal_order)
model_fit = await asyncio.to_thread(model.fit)

# Seed based on product_id to ensure deterministic wavy pattern matching historical std
np.random.seed(int(product_id) if product_id is not None else 42)
resid_std = float(np.std(model_fit.resid)) if hasattr(model_fit, "resid") else 10.0
noise = np.random.normal(0, resid_std * 0.4, size=len(forecast_res))

weekly_pattern = np.array([1.1 if d.weekday() in [4, 5] else 0.9 for d in forecast_res.index])
yhat_vals = (forecast_res.values * weekly_pattern + noise).tolist()
yhat_vals = [max(0.0, round(v)) for v in yhat_vals]
```